

## Project Design Phase-II

### Data Flow Diagram & User Stories

|               |                                                        |
|---------------|--------------------------------------------------------|
| Date          | 21 February 2026                                       |
| Team ID       | LTVIP2026TMIDS42560                                    |
| Project Name  | Online Payments Fraud Detection using Machine Learning |
| Maximum Marks | 4 Marks                                                |

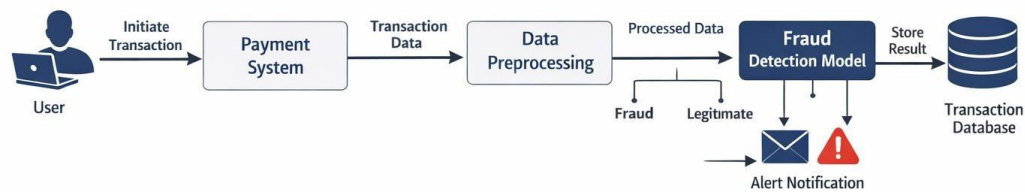
### Data Flow Diagram (DFD - Level 0)

| User Type       | Epic                 | User Story No. | User Story / Task                               | Acceptance Criteria                                         | Priority | Release  |
|-----------------|----------------------|----------------|-------------------------------------------------|-------------------------------------------------------------|----------|----------|
| Online Customer | Transaction Security | USN-1          | As a user, I can make online payments securely. | Transaction is classified as fraud or legitimate instantly. | High     | Sprint-1 |

|        |        |       |              |            |      |          |
|--------|--------|-------|--------------|------------|------|----------|
| Online | Alerts | USN-2 | As a user, I | Fraudulent | High | Sprint-1 |
|--------|--------|-------|--------------|------------|------|----------|

A Level-0 DFD illustrates how data flows between the user, payment system, preprocessing layer, machine learning fraud detection model, and transaction database.

### Online Payments Fraud Detection – Level 0 DFD



### User Stories

|               |            |       |                                                            |                                                            |        |          |
|---------------|------------|-------|------------------------------------------------------------|------------------------------------------------------------|--------|----------|
| Customer      |            |       | receive alerts for suspicious transactions.                | transactions trigger instant notification.                 |        |          |
| Fraud Analyst | Monitoring | USN-3 | As an analyst, I can view flagged fraudulent transactions. | Dashboard shows fraud probability and transaction details. | High   | Sprint-2 |
| Risk Manager  | Reporting  | USN-4 | As a manager, I can view fraud statistics and trends.      | System generates daily/weekly fraud reports.               | Medium | Sprint-2 |

|       |                  |       |                                                    |                                                             |        |          |
|-------|------------------|-------|----------------------------------------------------|-------------------------------------------------------------|--------|----------|
| Admin | Model Management | USN-5 | As an admin, I can update or retrain the ML model. | Updated model deploys without affecting system performance. | Medium | Sprint-3 |
|-------|------------------|-------|----------------------------------------------------|-------------------------------------------------------------|--------|----------|